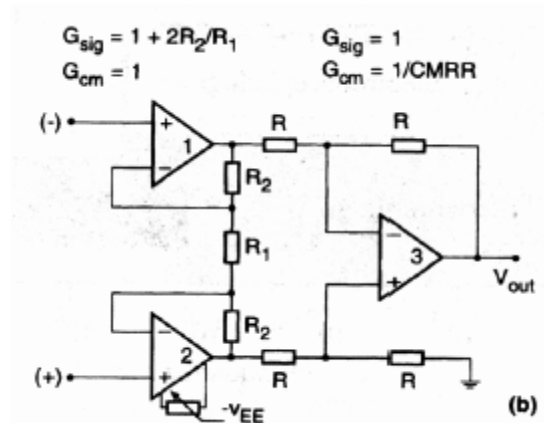


Hardware Project No. 1: Instrumentation Amplifier

The workhorse of almost any sensing application or interface is the instrumentation amplifier: a very clean, high-gain amplifier. A standard instrumentation amplifier is shown in the diagram below (taken from *The Biomedical Engineering Handbook*, a thick and weighty tome which touches on *many* subjects).



The circuit consists of three operational amplifiers and seven resistors (and an optional potentiometer). This is a two-stage amplifier; the input stage consists of two op amps and provides high differential gain, and unity common-mode gain. The output op amp is unity gain (differentially), but significantly reduces any common-mode signal. The gain of the entire amplifier is set by three resistors and given by:

$$K = 1 + \frac{2R_2}{R_1}$$

A potentiometer or resistor on one of the input stage op amps can provide offset trimming for the entire circuit.

Assignment

Build an instrumentation amplifier as shown in the above diagram. The amplifier should have a flat frequency response over its range of operation, which should be 5 Hz to 1 kHz. It should minimally have a gain of 200 or more (you should be able to get 1000 if you try; your main problem will be measuring such) over this range of frequencies. The amplifier will be powered by positive and negative power supplies, and should use a commonly used supply voltage, i.e., ± 6 , ± 9 or ± 12 V (note that an advantage of 9V, if you are able to use it, is that the circuit can be powered by two small batteries; however, that may limit your gain measurements, as your output may clip for a high gain; see below). You will need to measure:

- Frequency response (gain only, but you can provide phase information as well if you have access to a spectrum analyzer)
- Common-mode rejection ratio (measure either at 100 Hz or over a range of frequencies)
- Power consumption (you will need to measure the current drawn from each power supply)

For frequency response, a signal generator (possibly built-in to your 'scope) and oscilloscope can be used: you will need to make multiple measurements; starting at 5 Hz, then measure at 10, 20 and 50 Hz, and then multiples of 100. Measure past 1 kHz, until you see the gain decrease. Make your own

plot of the frequency response but also provide printouts of a selection (i.e., a few representative ones, not all of them) of the frequencies you measured. Note that using extremely low frequencies and amplitudes with a signal generator may not be the simplest task you have ever attempted. For example, a high gain means you need a very low input voltage to measure the output without clipping; maybe 15 V power supplies, or higher, are a good idea in this regard. Also, let us say you get a gain of 1,000. That means a 15 mV amplitude input will saturate your amplifier outputs, and your gain measurement will be inaccurate. Also, getting a 15 mV sine wave from a signal generator is...noisy, to say the least. It will be much better at producing a 150 mV amplitude, for example, signal. And you could knock that down to, say, the 10 mV you need it to be in order to measure accurately by using a precisely measured voltage divider. You may want to consider this trick. And please, *frequency response is conventionally plotted using a log frequency axis, with gain in dB.*

For emphasis: if your high-gain amplifier is not demonstrating the gain you expect it to, check your input and output signals! You may find your input is large enough such that the output is clipping.

The circuit can be built on a prototyping board of the type you have from your old Digital Logic Design kit, or you may get ambitious and wire wrap it, or put it on a printed circuit board. However you choose to build it, leave some room, as this circuit may be useful for another purpose later in the semester (a later project or two may benefit from an amplifier of this nature, though perhaps with a different gain). Be aware that it may take some time and effort to get decent matched resistor values. You are not required simulate the circuit ahead of time (using, say, SPICE), but doing so may lead to less time spent troubleshooting (or less time spent changing resistors afterward; perhaps it might be useful to examine the effect of resistors being mismatched here, *non?*) For example, simply writing in your report that *[insert my convenient excuse here]* may have been an issue in your gain not matching the design is a weak argument compared to writing it and providing a simulation that demonstrates it.

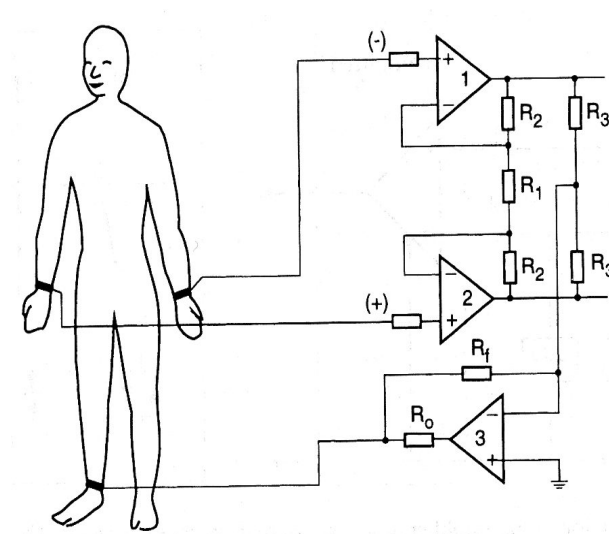
A report should include:

- Show me your knowledge of how op amps work. Using the circuit diagram and what we know about (ideal) op amps, derive the gain of the two stages of this circuit. You already know the answer, so that should help. Hint: you're looking for the differential output voltage of the first stage, which you can assume is everything to the left of the generic resistors labeled R. Then show unity gain for the second stage. Another hint: you do not need to artificially ground any inputs in order to derive this. Simple virtual-short-circuit-between-op-amp-terminals, and no-current-flows-into-the-op-amp-inputs will do it for you.
- A circuit diagram including component values, type of op amp used and any other pertinent information; you should include the desired (nominal) component values, as well as what you measured them to actually be. Also, it would be a good idea put the actual values into equations to see what your gain is.
- The above-mentioned information on frequency response, CMRR and power consumption
- A diagram or picture of the board layout
- A discussion of any difficulties you encountered in building the amplifier, and why it exceeds or does not meet any of the specifications
- If you do any simulation, include your SPICE simulation and data as an appendix (i.e., input files, output plots, what parameters you changed during/because of the simulation). Note that the default color scheme for plots (thin green lines on a black background) is very stylish on-screen, but really crappy on a piece of paper. *You* try to read it.

- The LM741 is a very, very lousy op amp. The LF411, on the other hand, is a fine, fine op amp. And anything that doesn't say 741 is usually better than anything that does.

Additional Task

We talked about interesting noise reduction techniques; let us add one to this circuit. The lecture notes discussed a “driven reference” circuit, and showed where such would plug into this amplifier. For your reference:



After building and testing your circuit, add this (resistors R3 to take the common-mode voltage, and the inverting op amp circuit). What should R3 be? How about “big enough” not to load your circuit. Try 1 M Ω ? Or 100 k Ω ? You are on the verge of becoming engineers, design something! If the point is to feed the inverted common-mode voltage as a reference, thus cancelling it out... The input to the inverting op amp is half the common-mode voltage, no? So maybe a gain of 2 for the amplifier? Do a quick test to be sure this doesn't severely affect the operation of your

amplifier (i.e., make sure the frequency response still looks similar), and we will fiddle around with this in class. If it does significantly affect the frequency response, disconnect the resistors R3 from your circuit, but be ready to reconnect them in class.

You will also be required to “bring the circuit to class” and demonstrate it to me (we will do this in the lab because there exists test equipment there and it's fun to be in a lab).

!!! Muy Importante !!!

While this is meant to be a biopotential amplifier, do not connect yourself to it in order to try it out and measure, say, your heart rate or the muscle activity in your arm (or some questionable amount of brain activity)! This circuit does not have any form of isolation to protect against the line voltage powering the oscilloscope or any power sources you may be using, and I do not wish to explain my role in your demise to your parents.

Due Date

February 26, 2026 (if you wish to hand a report in earlier, feel free to do so)

Show me the circuit. We will actually meet for class in the lab that night to demo the circuits. So, your other task is to find us appropriate lab space with scopes, power supplies, etc where we can meet and plug things in. This could be, for example, someone's senior project lab bench.

Afterwards, write the appropriate report and send it to me in the next week (so, by March 5).